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Policy Brief

Monopolization of the Satellite Broadband Market

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THE ISSUE

When war broke out in Ukraine last winter, the country's military became the newest customer of Elon Musk's Starlink program, which offers high-speed broadband service via satellites.[1] In wartime, the infrastructure of terrestrial fiber optic cables is vulnerable to air attacks and far more susceptible to interception or wiretapping than satellites' encrypted messaging. Ukraine's adoption of satellite communications exhibits the potential power of satellite technology, which could transform our communications systems.

But the booming satellite market is also full of pitfalls[2] brought on by the ongoing monopolization of the market. Just two companies – SpaceX's Starlink and Amazon's Project Kuiper – already have rapidly consolidated this new market, crowding out other competitors. Starlink owns 1,500 active satellites, amounting to near half of all satellites currently in low Earth orbit and is launching more every month at a breakneck pace.[3] Although Amazon lags behind Starlink, it acquired licenses from the Federal Communications Commission (FCC) two years ago to introduce a constellation of 3,000 satellites.[4]

These companies are jumping ahead of would-be rivals by evading adequate oversight, having received favorable treatment by regulatory agencies. With a dominant position in space, this satellite duopoly would aggravate the environmental hazard of space debris and worsen other consumer harms mainly through price increases and data surveillance.

Background on the Technology and Business of Satellites

Nanosatellites deliver fast Internet connection to remote or underserved parts of the globe by hovering far closer to Earth's surface than the conventional geostationary satellites used by governments and satellite TV providers like Dish.[5] Both Starlink and Amazon are using a new generation of nanosatellite technology that can provide faster internet service by flying close to Earth's surface. Their small size[6] makes them more cost-effective to mass produce.[7] Traveling in tightly knit mega constellations, the nanosatellites bounce radio transmissions between one another, creating a virtual grid across the band spectrum.[8] This technology has allowed Starlink and Project Kuiper to hit the threshold for high-speed broadband, notching 50Mbps to 150Mbps with low latency (a measure of the

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[1] Christopher Miller, Mark Scott, Bryan Bender, "UkraineX: How Elon Musk's Space Satelites Changed the War on the Ground," Politico, June 8, 2022.

[2] Luke Goldstein, "Why Are We Letting Monopolists Corner Space?" *Washington Monthly*, November 7, 2021.

[3] Erick Burgueño Salas, "Number of satellites in orbit by major country as of January 1, 2022," Statista, July 27, 2022.

[4] Elizabeth Howell, "The FCC has approved Amazon's plan for its Kuiper satellite constellation. Here's what that means," Space, August 20, 2020.

[5] Greg Ritchie and Thomas Seal, "Why Low-Earth Orbit Satellites Are the New Space Race," *Washington Post,* July 10, 2020.

[6] Chris Daehnick, Isabelle Klinghoffer, Ben Maritz, and Bill Wiseman, "Large LEO satellite constellations: Will it be different this time?," McKinsey, accessed September 28, 2022.

[7] Adrienne Harebottle, "Satellite Manufacturing in the Era of Mass Production," Satellite Today, August 19, 2021,

[8] CNBC, "Why Starlink Is Crucial to SpaceX's Success," YouTube, April 15, 2021.

lag time for wireless connection).[9] The companies are betting that if they keep improving the speed and lowering their prices they'll eventually be able to compete against fiber optic cable.[10]

Satellites have caught on with customers who either didn't have access to Internet connection in their areas before or were underserved by telecom giants. Starlink and Amazon are capitalizing on public frustration with telecom monopolies like AT&T, Verizon, and Comcast owing to unchecked monopoly power, which has led to higher prices in the U.S. for slower service compared with other industrialized nations.[11]

Starlink and Amazon are well aware of the market opportunity in satellites. Morgan Stanley estimates that the satellite communications sector alone will become a $400 billion business over the next two decades,[12] and Musk and Bezos want as big a piece of that action as they can get. In just the past few years, Starlink has drawn around 400,000 active users in 25 countries, mostly serving people in remote rural areas as well as a handful of companies such as airlines offering inflight WiFi.[13]

PROBLEMS

Starlink already owns around half of all satellites in low Earth orbit.[14] It's launching almost 120 per month[15] to build its current constellation of 12,000 – growing the total number of satellites that has ever been in the sky.[16] Musk is seeking the go-ahead for an additional 30,000 from U.S. and international regulators[17] to reach his stated goal of operating a constellation of 42,000.[18] Amazon isn't up and running yet but received approval for 3,000, which is almost as many as the total number of satellites currently in low Earth orbit.[19] Based on its beta tests, the company claims it can match, if not improve upon, the Internet speed of Starlink.[20] Both companies have jumped ahead of rivals by evading regulatory scrutiny and deploying monopolistic business practices.

1. Market Capture

On the business side, Musk has effectively engaged in predatory pricing to expand Starlink's customer base and capture market share. Customers who sign up for the Starlink program receive a kit that includes a user terminal to connect to the satellites, a mounting tripod, and a wireless router. According to its own disclosures, Starlink loses more than 50 percent on every kit it produces for customers to keep the price at $499 per package (and $99 per month for service).[21] This puts a huge strain on competitors but has worked in Starlink's favor. The company has added customers at a rapid rate, more than quadrupling a base of 90,000 in 2020 to 400,000 in 2022. While Project Kuiper hasn't set its rates

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[9] Bevin Fletcher, "Starlink speed competitiveness varies, as sign-ups surpass 500K," Fierce Wireless, May 5, 2021.

[10] "Elon Musk's SpaceX is launching the first of 12,000 Starlink satellites to cover Earth in high-speed internet. Here's how the ambitious project might work," Insider, May 15, 2019.

[11] John Cassidy, "We Need Real Competition, not a Cable-Internet Monopoly," *The New Yorker,* February 13, 2014.

[12] Michael Sheetz, "Morgan Stanley predicts space industry will triple in size: Here's how to invest," CNBC, October 12, 2017.

[13] Michael Sheetz, "SpaceX's Starlink satellite internet surpasses 400,000 subscribers globally," CNBC, May 25, 2022.

[14] Salas, "Number of satellites in orbit by major country as of January 1, 2022."

[15] Bojan Pancevski, "Elon Musk's Satellite Internet Project Is Too Risky, Rivals Say," *Wall Street Journal,* April 19, 2021.

[16] David Shepardson, "U.S. FCC approves SpaceX satellite deployment plan," Reuters, April 27, 2019.

[17] Jonathan O'Callaghan, "SpaceX's Application For 30,000 Extra Starlink Satellites Highlights Concerns About Regulation," *Forbes,* Oct 16, 2019.

[18] Caleb Henry, "SpaceX submits paperwork for 30,000 more Starlink satellites," Space News, October 15, 2019.

[19] Elizabeth Howell, "The FCC has approved Amazon's plan for its Kuiper satellite constellation. Here's what that means," Space, August 20, 2020.

[20] Jon Brodkin, "Amazon's answer to SpaceX Starlink delivers 400Mbps in prototype phase," ArsTechnica, December 17, 2020.

[21] Aria Alamalhodaei, "SpaceX is losing money on its Starlink terminals, but sees lower costs ahead," TechCrunch, June 29, 2021.

yet, the company is expected to match Starlink's prices once it gets its fleet up and running. Such tactics kill competition and in the long term could lead to astronomically higher pricing once the duopoly has eliminated all potential competition and is in full control of the market.

2. Data Collection

As the industry centralizes around SpaceX and Amazon, data-gathering and surveillance are becoming major concerns. In 2017, President Donald Trump signed a bill revoking a set of Obama-era privacy protections that restricted telecommunications companies from collecting web browser search history, location tracking, and other data from its customers. By rescinding those protections, the bill opened the floodgates to selling personal information to third parties. The same rules apply for satellite communications.

As the satellite communications sector attracts a broader customer base, Starlink and Project Kuiper will control even more of its customers' personal information. The companies could monetize data collection to build more advanced business models that would help crush smaller satellite competitors. They can also sell that data for further profit at the cost of user privacy as Google and Facebook have done. Amazon already has a robust data collection business in its retail sector, which the company could pair with telecom data to become a truly terrifying micro-targeting data empire.

3. Regulatory Negligence

Starlink has managed to secure its dominant position in space by wielding power and influence over agencies. The two most important American regulatory agencies for overseeing satellite commerce are the FCC and the Federal Aviation Administration.

For companies to get satellites into space, they first must go through the FCC's auctioning process for band spectrum licenses. The broadband spectrum where radio transmissions travel is the equivalent of atmospheric real estate: only so much valuable land exists.[22] To receive regulatory approval, companies first need to purchase licensing for both the atmospheric real estate and the allotted band spectrum "blocks" for radio transmissions, which has become a hot market.[23] SpaceX has quickly bought up[24] these licensing rights and crowded out other competitors by blocking them from trespassing in its spectrum.[25]

After getting licensing rights, companies then need to file application requests to the FCC for satellite launches. Since 2018, the FCC routinely has signed off on both Starlink's and Amazon's applications to send out massive satellite fleets without adequate oversight. Similar to issuing government contracts, satellite licensing requires a review process as multiple companies typically want access to the best positioned band spectrum in space.

When it comes to Starlink, the agency has not properly accounted for collision rate or even made sure that the companies seeking approval offer the best service, according to multiple watchdog groups.[26] For instance, Starlink reported malfunctions in around 5% of its satellites in 2019 due to technical flaws.[27] Although there's no standardized metric for satellite failures, experts argue that the massive scale of Starlink's constellation will magnify the damaging effects of its satellite failures. Rivals have also pointed to customer dissatisfaction with Starlink's service,[28] claiming that it isn't as fast as

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[22] Federal Communications Commission Licensing.https://www.fcc.gov/licensing

[23] Federal Communications Commission Auctions.

[24] "Coalition Pushes FCC to Open 12 GHz While Incumbents Say 'No'," Inside Towers, July 9, 2021.

[25] Monica Alleven, "Dish, RS Access cheer FCC move to re-evaluate 12 GHz band," Fierce Wireless, January 13, 2021.

[26] Bojan Pancevski, "Elon Musk's Satellite Internet Project Is Too Risky, Rivals Say," *Wall Street Journal,* April 19, 2021.

[27] Jonathan O'Callaghan, "'Not Good Enough' - SpaceX Reveals That 5% Of Its Starlink Satellites Have Failed In Orbit So Far," *Forbes*, June 30, 2019.

[28] Sara Morrison, "The FCC's big bet on Elon Musk," Vox, May 17, 2021.

advertised.[29] None of these complaints has apparently given the FCC pause in approving Starlink's fleets.

When it comes to Starlink, government agencies have plied the company with federal grants, subsidies, and lucrative contracts.[30] The U.S. Air Force in August signed a deal with Starlink, making it the exclusive Internet provider in many countries where it operates.[31] Recently, the FCC was forced to cancel a billion-dollar grant awarded to Starlink[32] after investigative reporting showed Starlink had fudged numbers on its application.[33] The incident is a stark example of how Starlink has repeatedly tried to exploit agency negligence.

4. Environmental Effects

By approving vast fleets of Starlink and Amazon's satellites, the FCC has unleashed several dangers for space management. For one, by flying at such low levels in the Earth's orbit,[34] Starlink is creating light pollution that interferes with astronomers' ability to study space.[35] SpaceX has tried applying a black coating to minimize the rays of light refracted off its satellites, but astronomers don't think it will have much effect.[36]

Even more concerning is the threat of space debris, which already has turned into a full-blown crisis. Approximately 30,000 space junk parts have accumulated in Earth's orbit, careening haphazardly through the atmosphere.[37] Because these particles travel at thousands of miles per hour, even a minor impact between an active satellite and a tiny scrap of shrapnel is dangerous. In a worst-case scenario, even a minor collision could create a cascading effect that would block off huge swaths of Earth's orbit for space travel.[38]

Recent incidents have made these fears seem far less speculative. Just two satellite incidents – one collision involving an inactive Russian satellite in 2009 and another precipitated by a Chinese anti-satellite missile test in 2007 – account for a significant portion of the debris currently in orbit, which is likely to remain there for decades.[39]

Even outside of these disaster scenarios, space debris could cause serious problems. It could knock out key parts of the communications infrastructure, as well as other intelligence satellites owned by the U.S. government.

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[29] Rachel Kraus, "New Speedtest data show Starlink internet speeds and latency are all over the place," Mashable, May 5, 2021.
[30] Ryan Tracy, "Elon Musk's SpaceX Riles Its Rivals for Broadband Subsidies," *Wall Street Journal*, January 31, 2021.
[31] Sandra Erwin, "SpaceX gets $1.9 million Air Force contract for Starlink services in Europe and Africa," Space News, August 15, 2022.
[32] Michael Sheetz, "FCC denies SpaceX bid for nearly $1 billion in rural broadband subsidies for Starlink," CNBC, August 10, 2022.
[33] S. Derek Turner, "Broadband Boondoggle: Ajit Pai's $886M Gift to Elon Musk," Free Press, December 14, 2020.
[34] Greg Ritchie and Thomas Seal, "Why Low-Earth Orbit Satellites Are the New Space Race," Bloomberg, July 10, 2020.
[35] Brian Resnick, "The night sky is increasingly dystopian," Vox, January 29, 2020.
[36] Jon Brodkin, "SpaceX satellites' effect on night sky can't be eliminated, astronomers say," Ars Technica, August 29, 2020.
[37] NASA, "Space Debris and Human Spacecraft," May 25, 2021.
[38] Leonard David, "China's Anti-Satellite Test: Worrisome Debris Cloud Circles Earth," Space, February 2, 2007.
[39] Brian Weeden, "2009 Iridium-Cosmos Collision Fact Sheet," Secure World Foundation, November 10, 2020.

SOLUTIONS

As the situation in Ukraine demonstrates, satellite technology provides beneficial applications. Satellites could help bridge the digital divide, extending Internet service to underserved areas. But to achieve that, we need a more robust regulatory framework and public policy to structure the commercial competition in space so that it is both fair and safe.

When new communications technologies emerged in the past, the U.S. government responded with legislation for allocating key assets and managing competition. Radio technology has echoes of the current dilemma with satellite technology. In the 1920s, applications for new radio licenses came in at such a pace that the Commerce Department started approving licenses to almost anyone who applied.[40] Given limited radio spectrum, this created massive problems with radio interference.[41] Congress responded with the Radio Act in 1927, which set more stringent criteria for reviewing licensing applications.[42] Rather than give them out on a first-come, first-served basis, station licenses went to applicants who could best demonstrate their ability to serve the public as stewards of a radio spectrum commonweal.[43]

Later legislation further managed competition by putting brakes on any tendency toward monopoly. Until many of these strictures were repealed in the 1980s and '90s, this meant limiting the number of stations any one person or corporation could control, prohibiting station owners from vertically integrating into ownership of other forms of media, and even requiring that radio stations devote a share of their airwaves to balanced public interest programming.

Today, we need a new satellite communications bill of similar scale and scope. One example would be to ban satellite telecom companies from selling customer data. The FCC should also be pushed to slow down satellite launches to adequately review licensing applications from companies. As with the original Ma Bell telephone system, companies that own essential communications technology shouldn't be able to use this access to information to get into other lines of business; Amazon shouldn't be allowed to use data it has gathered from its ecommerce or entertainment platform to advance its satellite communications operations and vice versa.

To deal with the space debris crisis, we'll also need international cooperation. The U.S. should work with other countries to set up an international space traffic system with patrol officers who manage overcrowding.[44]

For more information, contact:
General media and policy contact: info@openmarketsinstitute.org
Luke Goldstein, reporter/researcher, Center for Journalism & Liberty, lgoldstein@openmarketsinstitute.org
Phillip Longman, policy director, Open Markets Institute, longman@openmarketsinstitute.org

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[40] Communications Act of 1934, Britannica.
[41] Ibid.
[42] Ibid.
[43] Ibid.
[44] U.S. Senate Committee on Commerce, Science, and Transportation, "Space Missions of Global Importance: Planetary Defense, Space Weather Protection, and Space Situational Awareness," February 12, 2020.